# JAFFEDRONE

**A drone instrument and meditation tool for the Miyoo Mini Plus**

User manual · version 0.8.1




github.com/sydilaxe/Jaffedrone-dev

# Getting started




Jaffedrone is a drone of five voices: strings that are bowed and plucked, or wind voices that are blown and tongued, tuned to a scale and moving through chords. Each voice has its own rhythm, or you play it by hand.

## Install

Unzip Jaffedrone-miyoo.zip at the root of the SD card (it holds App/Jaffedrone/) and open Apps → Jaffedrone. The start-up screen shows for a moment; any button moves on.

## The first buttons

| | | |
|---|---|---|
| 1 | START | start or stop the drone (it fades in and out) |
| 2 | START + L1 / R1 | previous / next page: PLAY, SOUND, COMPOSE, MIX, SONG |
| | SELECT | the next chord |
| 3 | MENU | leave the app |

Listen on headphones for the deepest low end and for binaural beats; the speaker plays little below about 200 Hz.

## The five pages



PLAY: the strings and their rhythms



SOUND: each voice's tone



COMPOSE: scale, chords, change



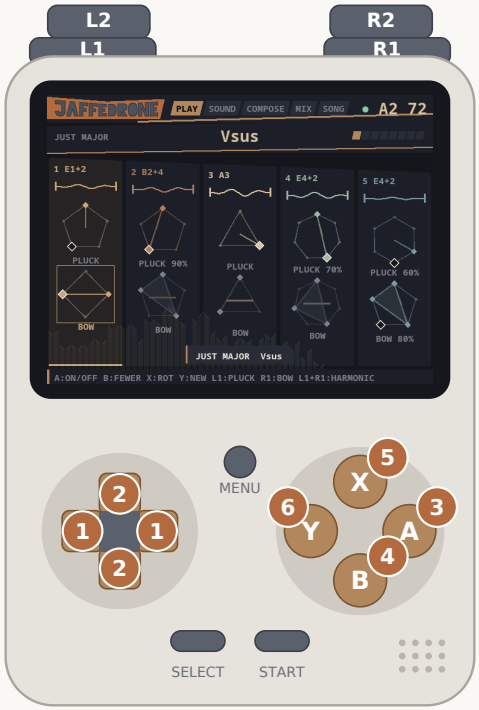
MIX: levels and effects



SONG: presets, chord memory, key

START + R1 moves right along the tabs at the top, START + L1 left. On SOUND, COMPOSE and MIX, L1 and R1 alone change page too.

# PLAY: the rhythm rings




Each string has two Euclidean rings: the pluck ring plucks on its pulses and the bow ring swells the bow on its pulses (off, the bow plays steadily). A ring spreads its pulses as evenly as it can over its steps, like the rhythms of HI5 and RadioScan.

| | | |
|---|---|---|
| 1 | ◄► | choose a string |
| 2 | ▲▼ | choose its pluck ring or bow ring |
| 3 | A | switch the ring on or off |
| 4 | B | fewer pulses, wrapping back to all |
| 5 | X | rotate the pattern |
| 6 | Y | a new simple pattern |

A label like PLUCK 70% means each pulse of that ring plays seven times in ten (set with R2 + X / Y).

## Shaping a ring: hold L2




| | | |
|---|---|---|
| 1 | hold L2 | then: |
| 2 | ▲▼ | more / fewer steps |
| 3 | ◄► | step length (right is faster): 1/16 to 2 bars |
| 4 | X | more pulses |
| 5 | Y | rotate back |
| 6 | A | mute or unmute the string (a red bar marks it) |
| | tap L2 | the quiet view (see The quiet view) |

# Playing by hand

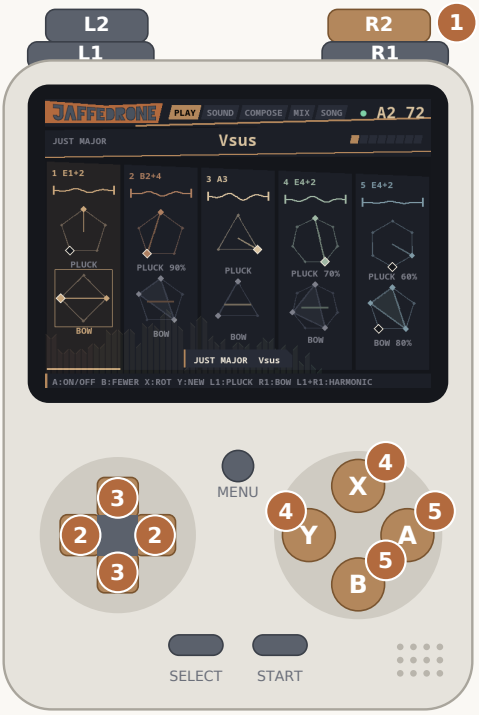
The shoulder buttons play the string selected on PLAY, over whatever its rings are doing. A wind voice is tongued and blown instead.




| | | |
|---|---|---|
| 1 | L1 | pluck the selected string (wind: tongue a note) |
| 2 | R1 held | bow it: the bow is drawn in slowly, swelling over about two seconds and brightening as it bites, and sings while you hold (wind: blow) |
| | hold L1, then R1 | bow with a light touch at a node: the next harmonic up sings (wind: overblow, an octave up, a twelfth for the reed) |

The bowed string's column lights while the bow is on it.

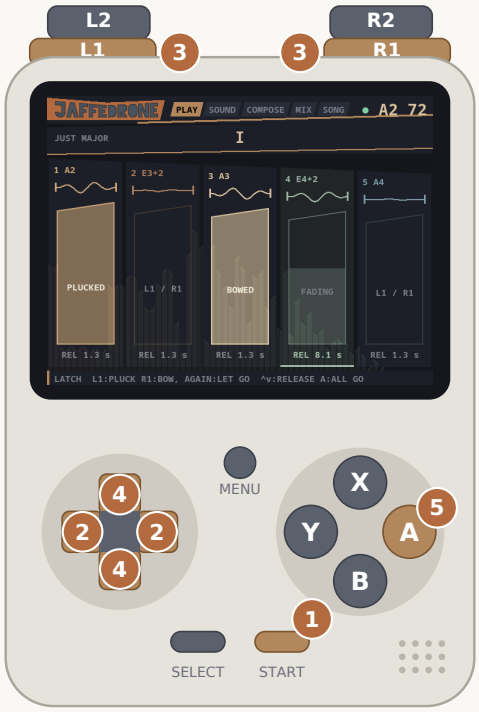
## Quick sound changes: hold R2




| | | |
|---|---|---|
| 1 | hold R2 | then: |
| 2 | ◄ ► | previous / next scale |
| 3 | ▲ ▼ | tone: brighter / darker |
| 4 | X / Y | the selected ring's chance, more / less |
| 5 | A / B | body resonance, more / less |

# LATCH: hold each string yourself

In LATCH the rings rest and every string is silent until you play it. A string you pluck or bow keeps droning, with no need to hold the button, until you play it again; then it fades over its own RELEASE.




| | | |
|---|---|---|
| **1** | START + A | switch PLAY MODE between RINGS and LATCH (also on the SONG page) |
| **2** | ◄► | choose a string |
| **3** | L1 / R1 | pluck / bow it on: it drones. Either again: it lets go and fades |
| **4** | ▲▼ | that string's RELEASE, 0.05 s to 20 s |
| **5** | A | let every string go |

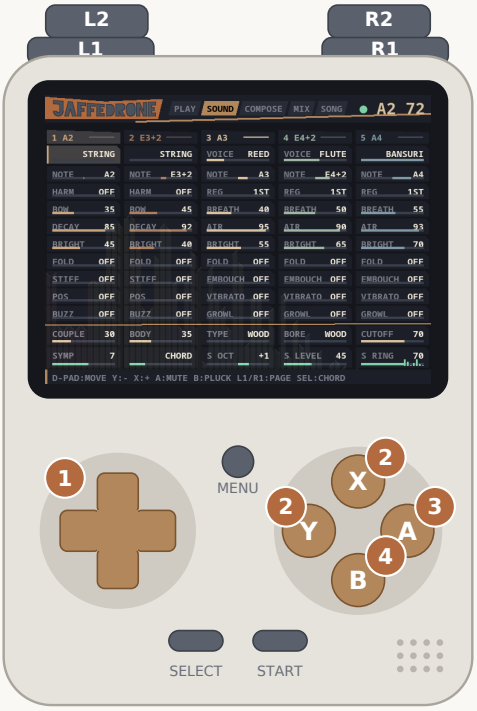
Each column shows a board with a shed-roof top. It fills while its string drones and drains as it fades, marked PLUCKED, BOWED or FADING, with the RELEASE beneath. A pluck starts at once; a bow is drawn in slowly. The RELEASE times are also on the MIX page, and LATCH is saved with songs and presets.

## Try it

Press START + A. Bow string 1 with R1 for a low drone, move to string 3 and pluck it with L1, then press L1 on string 3 again to let it ring away. Raise string 5's RELEASE with ▲ to ten seconds or more for a long dying shimmer.

# SOUND, COMPOSE and MIX

On these pages the d-pad moves to the nearest control in that direction, and Y / X lower / raise it. Held, they speed up.




| | | |
|---|---|---|
| 1 | d-pad | move between controls |
| 2 | Y / X | lower / raise the control |
| 3 | A | on a string's control: mute or unmute that string |
| 4 | B | pluck the string (to hear a change) |

## SOUND

The VOICE row makes each column a STRING or one of seven winds: LIP (brass, conch), REED (clarinet), DOUBLE (oboe, shehnai), FLUTE, FREE (harmonium), RECORDER or BANSURI. The rows below follow the voice: a string has BOW, DECAY, BRIGHT, FOLD, STIFF, POS and BUZZ; a wind has BREATH, AIR, EMBOUCH, VIBRATO and GROWL. The shared row holds COUPLE, BODY, its TYPE, the winds' BORE and CUTOFF; the last row the sympathetic strings.




COMPOSE: the scale and root (a wheel of the scale's notes, the chord lit), how often chords change and how: MOTION, HOME, COLOR, CHANCE, RELATE, EVOLVE, GLIDE, tempo, Wi-Fi sync and two LFOs.




MIX: each voice's LEVEL, PAN and GATE (RELEASE in LATCH), the master, undertones (SUB), then delay, reverb and binaural.

# SONG: presets, chord memory and key




| | | |
|---|---|---|
| **1** | d-pad | choose a preset slot or a setting |
| **2** | A | load the preset (your Wi-Fi setting stays) |
| **3** | X | save the sound and rhythms to the slot |
| **4** | R2 + Y | clear the slot, or the chord memory |

Ten preset slots keep everything: voices, rhythms, scale, effects, the chord memory and PLAY MODE.

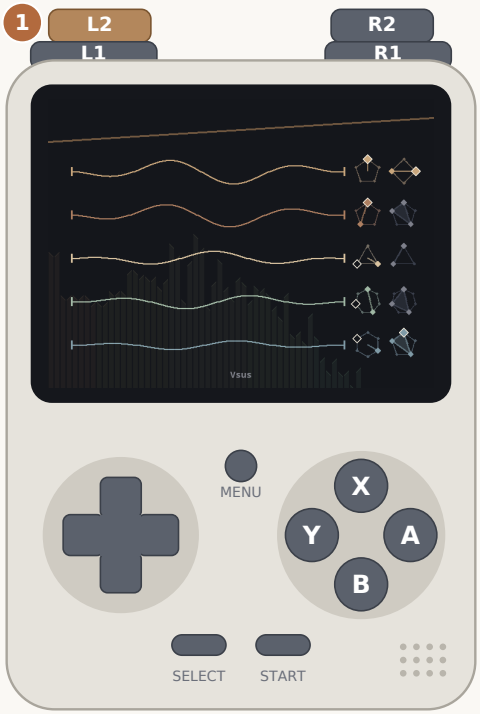
## Chord memory

Set PROG to RECORD and every chord change (SELECT or CHANGES) is kept, up to 16. PLAY steps through them round and round; FREE lets the algorithm choose again.

## Playing with others

KEY LOCK holds the root, scale and MODE (the mode chords come from in EQUAL) and tunes chords exactly. ATTACK and RELEASE set how the drone fades in and out with START.

# The quiet view and the projection




Tap L2 on PLAY for the quiet view: the five strings across the screen with their rings, and only the chord in words. Tap again to come back.

| | | |
|---|---|---|
| 1 | tap L2 | the quiet view on / off |




Behind every page a faint projection glows, like the light across the Jaffe house's ceiling: the sound's spectrum as siding boards, low notes on the left, high on the right, each pair's tops sloping into a roof valley. With binaural on, two projector beams fall from the top corners and trade left and right at the beat.

| | | |
|---|---|---|
| 1 | START + Y | projection SOFT, BRIGHT or OFF (remembered) |

# Binaural, ombak and tunings

On MIX, BEAT sets a binaural beat from 0.1 to 40 Hz: the voices below about 1 kHz are shifted down in the left ear and up in the right, so every low partial beats between the ears. DEPTH sets how much. MODE:

| | |
|---|---|
| **FREE** | the beat as set |
| **TEMPO** | the beat moves to a power-of-two multiple of the tempo |
| **A>T 10M** | a journey from alpha (10 Hz) down to theta (6 Hz) over ten minutes |
| **T>D 15M** | from theta (6 Hz) down to delta (2.5 Hz) over fifteen minutes |
| **OMBAK** | the beat for both ears, like a Balinese gamelan's paired instruments: the low end swells and fades, and it works on the speaker |

## Settings to start from

| | |
|---|---|
| **Meditation** | BEAT 6 MEDITATE, headphones; or MODE A>T 10M to drift down from alpha |
| **Sleep** | MODE T>D 15M, headphones, a long ATTACK and RELEASE |
| **Gamelan shimmer** | scale TIRUS or BEGBEG, MODE OMBAK, BEAT 7 Hz, on the speaker |

The research on binaural beats is limited and mixed; these are the settings most used, not a promise of an effect.

## Tunings

COMPOSE, or R2 + ◄ / ► on PLAY, chooses the scale: EQUAL (with its modes), just major and minor, the harmonic series, just pentatonics, the ragas YAMAN, BHAIRAV, KAFI, BHUPALI and MALKAUNS, the gamelan tunings SLENDRO, PELOG and SELISIR, and the Balinese gong kebyar tunings BEGBEG, SEDANG and TIRUS (Toth's profiles, as reproduced by Vitale and Sethares). Each scale has its own way of moving through chords.

# Quick reference




| | |
|---|---|
| START | start / stop the drone |
| START + L1/R1 | previous / next page |
| START + A | PLAY MODE: RINGS / LATCH |
| START + Y | projection SOFT / BRIGHT / OFF |
| START + SELECT | the log: OFF / BASIC / DETAIL |
| SELECT | next chord |
| L1 | pluck (wind: tongue); LATCH: pluck on / let go |
| R1 | bow while held (wind: blow); LATCH: bow on / let go |
| L1 **then** R1 | harmonic (wind: overblow) |
| **tap** L2 | quiet view |
| L2 **+ ...** | shape the ring, mute (A) |
| R2 **+ ...** | scale, tone, chance, body |
| MENU | leave |

## The log

START + SELECT cycles the log through OFF, BASIC and DETAIL. It is written to App/Jaffedrone/jaffedrone_log.txt: how hard the audio is working, any dropouts, the levels, each voice's tuning (DETAIL) and what was played, with suggestions when something could be better. If you hear crackles, set JD_BUFFER=2048 in App/Jaffedrone/launch.sh.